

Chapter 5. Carbon dioxide in occupied classrooms: one spring term in six Brackenfirth primary and infant schools

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Summary of the chapter

This chapter takes up something that the plant-room chapters of this thesis cannot settle: do the children in the estate's classrooms get enough fresh air while they are being taught? Carbon dioxide (CO₂) loggers were placed in 36 classrooms of six primary and infant schools of the Brackenfirth estate for the spring term of 2026. Over the whole term, the CO₂ concentration was above 1,500 ppm for 17% of teaching hours. The figure hides a wide spread: classrooms with a mechanical air supply were above the level for a tenth of the time, classrooms ventilated only by windows for more than three times as long, and one school was far worse than the other five. The chapter describes the measurements, reports them by school and by type of ventilation, and ends with what the teachers told us and with one small practical experiment.

5.1 Why measure the classrooms

Chapters 3 and 4 look at buildings from the plant room: Chapter 4 describes my part in the laboratory's fault detector for air-handling plant. A plant room can be in perfect order while a classroom at the end of a long duct, or a classroom with no duct at all, is stuffy by mid-morning. Carbon dioxide breathed out by the occupants is the usual marker for this. It is not harmful at the concentrations found in schools, but it rises and falls with the amount of outdoor air each person receives, and several studies link high classroom concentrations to drowsiness and to poorer scores on attention tasks.

The national Schools Ventilation Guidance Note 4 (SVG-4) sets the reference levels used here. For a classroom ventilated by opening windows, the mean concentration over the teaching day should not exceed 1,500 ppm. For a classroom with a mechanical supply the design target is a teaching-day mean of 1,000 ppm. In either kind of room the concentration should not stay above 2,000 ppm for more than 20 consecutive minutes. To compare all rooms on one scale, this chapter uses 1,500 ppm throughout as the level of concern, and reports the stricter and looser levels only in passing.

5.2 Design

Schools. Nine primary and infant schools of the estate were approached through the council's estates office in October 2025, and the head teachers of six agreed to take part. The schools are therefore not a random sample; they are the ones whose heads were willing. They are referred to by letter throughout this chapter, from the largest pupil roll (A) to the smallest (F). The correspondence between the letters and the estate's building codes is kept in the logger register and is not repeated here. No secondary school took part.

Classrooms. In each school the head teacher and I chose the rooms together, aiming for one room per year group and for every room without a ducted supply. This gave 36 classrooms: eight in School A, seven in School B, six each in Schools C and D, five in School E and four in School F. Of the 36, 26 have a mechanical air supply and ten are ventilated by windows alone. The window-only rooms are in annexes, temporary buildings and older wings; four of the six rooms logged in School C, all of its upper-school rooms, are of this kind.

Loggers. One type of logger was used throughout; it is described in section 5.3. Before any logger went into a school, all of them were run side by side in the laboratory for two days against a certified 1,000 ppm reference gas, and every one read within 20 ppm of it. The same gas was used again at the end of the study (Appendix 5.A).

Trial of the procedure. Before the main study the procedure was tried in two classrooms of School B for two weeks from 17 November 2025. The trial was about practicalities: where a logger survives a school day, how often it gets unplugged, and how long a download takes. In those two rooms, both of which have a mechanical supply, the median concentration in teaching hours was 1,240 ppm. The teacher questionnaire described in section 5.6 was piloted at the same time with 8 teachers at School B, and two of its items were reworded as a result.

Period. The main study ran from Monday 12 January to Friday 27 March 2026, the whole of the spring term. The half-term week of 16 to 20 February is excluded. That leaves ten school weeks and 50 school days, and so 1,800 possible classroom-days.

5.3 Instruments and data handling

Each classroom had one Sentrel Q2 logger, a non-dispersive infrared instrument that also records air temperature and relative humidity. The maker states an accuracy of ± 50 ppm plus 3% of the reading. The loggers recorded at 2-minute intervals and ran on mains power with a battery reserve. Each was fixed between 1.0 and 1.4 metres above the floor, on an interior wall or shelf, at least two metres from any window, door or supply grille and out of the breathing zone of the nearest desk.

The loggers were installed on 8 and 9 January, in the week before term, and collected on 1 and 2 April. One further logger of the same type was mounted on the roof of School A as an outdoor reference; its mean over the term was 428 ppm.

Teaching hours are taken as 8.50 am to 3.20 pm on school days, the timetable common to all six schools, with no adjustment for lunch or assemblies. A classroom-day counts as valid when at least 85% of its teaching-hours readings are present and the logger was in its fixed position. Days on which a room was closed, a class was away on a trip, or a logger had been moved or unplugged were dropped. Of the 1,800 possible classroom-days, 1,692 are valid, which is 94%.

5.4 Measures

Three measures are used. The first, and the one quoted in the summary, is the share of valid teaching hours with the concentration above 1,500 ppm; shares for a group of rooms are pooled over all their valid readings and are not averages of room averages. The second is the number of classroom-days whose teaching-day mean is above 1,500 ppm, which is the form in which SVG-4 states its limit. The third is the median and the 95th percentile of the concentration in teaching hours.

5.5 Results

5.5.1 All classrooms

Across the 36 classrooms the concentration was above 1,500 ppm for 17% of valid teaching hours. The median concentration in teaching hours was 1,150 ppm and the 95th percentile 2,480 ppm. On 203 of the 1,692 valid classroom-days, which is 12%, the teaching-day mean was above 1,500 ppm. Against the stricter level of 1,000 ppm the picture is less comfortable: the concentration was above it for well over half of all teaching hours.

Exceedances were concentrated in the colder first half of the term, when windows were opened least. No figure for the two halves of the term is given here, because the two loggers that failed the check at retrieval lost their data from the second half only, and a like-for-like split has not yet been computed.

5.5.2 By school

Table 5.2. Concentration in teaching hours by school, spring term 2026.

School	Classrooms logged	Share of teaching hours above 1,500 ppm	Median (ppm)
A	8	11%	1,080
B	7	14%	1,110
C	6	33%	1,430
D	6	18%	1,160
E	5	9%	1,060
F	4	16%	1,140

School C stands apart. Its classrooms were above 1,500 ppm for 33% of teaching hours, three times the share in School A, and its median of 1,430 ppm is close to the level of concern itself. School E, an infants' school with small classes and a mechanical supply to every logged room, had the lowest share, 9%. The ranking by median is the same as the ranking by share.

5.5.3 By type of ventilation

The difference between schools is largely a difference between kinds of room. In the 26 classrooms with a mechanical supply the concentration was above 1,500 ppm for 10% of teaching hours. In the ten window-only classrooms it was above that level for 35%. School C has four of the ten window-only rooms, School B, School D and School F two each. Within School C, the two rooms that do have a mechanical supply look like mechanically supplied rooms elsewhere.

5.5.4 Through the day

In window-only rooms the concentration climbed through the morning, was highest in the hour before lunch and fell back when the room emptied at midday. In rooms with a mechanical supply the daily curve was flatter and peaked in the last lesson of the afternoon. Mondays were no different from other days.

5.5.5 Excursions above 2,000 ppm

SVG-4 also asks that the concentration should not stay above 2,000 ppm for more than 20 consecutive minutes. That happened at least once during the term in 21 of the 36 classrooms, and in every one of the ten window-only rooms. In the window-only rooms it was a routine event, on roughly two school days in five; in the rooms with a mechanical supply it was occasional and usually coincided with a wet lunchtime, when the class stayed indoors.

5.5.6 Year groups, room size and temperature

Older year groups had higher readings than younger ones in the same school, as would be expected from larger children in rooms of the same size. The difference between Reception and Year 6 rooms with a mechanical supply was modest, a median of about 1,030 ppm against about 1,140 ppm. Floor area, which varies little between the logged rooms, explained nothing once the type of ventilation was taken into account.

The window-only rooms were also the colder rooms. Their mean air temperature in teaching hours was 19.1 °C, against 20.6 °C in the rooms with a mechanical supply, and on the coldest mornings several of them were below 17 °C at the start of the first lesson. This matters for the argument of section 5.6: the teachers who keep their windows shut are not being careless, they are choosing between two discomforts. Relative humidity stayed between 38% and 58% in every room for nearly all of the term and is not discussed further.

5.6 What the teachers said

A two-page questionnaire went, in the last fortnight of term, to every member of staff who had taught in a logged room during the term, including job-share partners and long-term cover teachers. It was returned by 33 teachers. Asked why they kept windows shut on cold mornings, 19 of the 33 named draughts on the children sitting nearest the window, 9 named noise from the road or the playground, and the rest gave other reasons or none. Only six said they had ever been told what a healthy CO₂ level is. No teacher in a room with a mechanical supply knew whether the supply could be adjusted from the room.

5.7 Indicator lights in six rooms

For the last three school weeks, from 9 March, the six window-only classrooms in Schools C and D were each given a small desk-top indicator that shows green below 1,200 ppm, amber above it and red above 1,500 ppm. Nothing else was changed and the teachers were asked only to open a window when they saw red. In those six rooms the concentration had been above 1,500 ppm for 45% of teaching hours in the seven weeks before the lights arrived. In the three weeks with the lights it was above that level for 27% of teaching hours. The four window-only rooms in Schools B and F, which had no lights, were not analysed separately for the two periods, so this is a before-and-after observation and not a controlled comparison. March was also milder than January, and milder weather alone opens windows.

5.8 Limitations

The study covers one term, in one winter, in six schools that volunteered. A single logger per room measures one point, and rooms are not perfectly mixed. The loggers were visible to the class, and some teachers may have opened windows more often simply because an instrument was on the wall. No count of the pupils present was kept, so nothing here can be expressed per pupil. The readings were not related to the condition of the air-handling plant on the same days; that comparison is outside this chapter. The indicator-light experiment has no comparison group and coincides with warmer weather.

5.9 Conclusions for the estate

Most classrooms with a mechanical supply meet the level of concern most of the time. The window-only rooms do not, and the estate has more of them in teaching use than its drawings suggest. A cheap indicator may help, but the evidence for it here is weak, and it should be tried properly through a whole winter before the estate buys any. School-level results were sent to the six head teachers in July, ahead of this chapter, as agreed when they joined.

Appendix 5.A Logger checks at retrieval

On collection every logger was again exposed to the 1,000 ppm reference gas. A logger passed if it read within 75 ppm of the reference. All but two of the 36 loggers passed; among those that passed, the median deviation was 23 ppm and the largest 49 ppm. Two loggers failed, one in School E reading 96 ppm high and one in School F reading 88 ppm high. For each of these the readings were compared with the neighbouring classroom's logger to find when the two began to part, and the days after that point were

removed: twelve school days for the logger in School E and eleven for the one in School F. Those removals are included in the count of valid classroom-days in section 5.3.